

Hoisting Javascript

What is Hoisting in JavaScript?

Hoisting is JavaScript's default behavior of moving declarations (not values) to the top of their scope (global or function scope) before code execution.

- This means you can sometimes use variables or functions **before they are declared** in the code.

1. Hoisting with var

Variables declared with var are **hoisted and initialized as undefined**.

Example:

```
console.log(a); // undefined
```

```
var a = 10;
```

How JS sees it internally:

```
var a;
```

```
console.log(a);
```

```
a = 10;
```

✓ No error

✗ Value is undefined

- **2. Hoisting with let and const**

let and const are **hoisted** but **NOT initialized**.

```
console.log(b); // ❌ ReferenceError
```

```
let b = 20;
```

Temporal Dead Zone (TDZ)

The time between **hoisting** and **initialization** is called the **Temporal Dead Zone**.

Example:

```
{  
  console.log(x); // ❌ TDZ  
  let x = 5;  
}
```

3. Hoisting with Function Declarations

Function declarations are **fully hoisted** (both name and body).

```
sayHello();
```

```
function sayHello() {  
  console.log("Hello!");  
}
```

✓ Works perfectly

4. Function Expressions and Hoisting

Using var

sayHi(); // ❌ TypeError

```
var sayHi = function () {  
  console.log("Hi");  
};
```

Reason:

var sayHi is hoisted as undefined, not the function.

Using let or const

sayHey(); // ❌ ReferenceError

```
const sayHey = () => {  
  console.log("Hey");  
};
```

Not usable before initialization.

5. Arrow Functions and Hoisting

Arrow functions behave like function expressions.

```
greet(); // ❌ ReferenceError
```

```
const greet = () => {  
  console.log("Hello");  
};
```

❌ Not hoisted like function declarations.

6. Class Hoisting

Classes are also hoisted **but in TDZ**.

```
const obj = new Person(); // ❌ ReferenceError
```

```
class Person {  
  constructor(name) {  
    this.name = name;  
  }  
}
```

Declaration Type	Hoisted	Initialized	Can Use Before Declaration
7. Summary Table			
var	✓	undefined	✓ (but unsafe)
let	✓	✗	✗
const	✓	✗	✗
Function Declaration	✓	✓	✓
Function Expression	✓ (var only)	✗	✗
Arrow Function	✗	✗	✗
Class	✓	✗ (TDZ)	✗